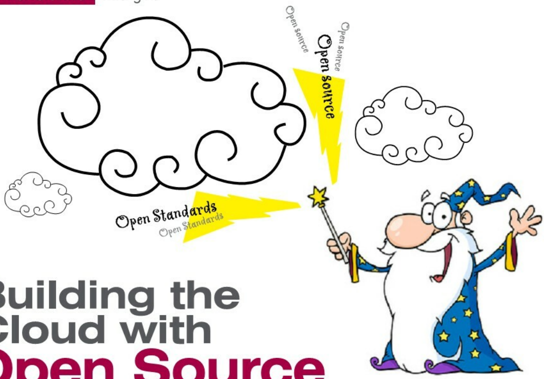


# Building the Cloud with Open Source and Open Standards

Open source software and standards are not just beneficial, but highly essential for a heterogeneous, shared and scalable environment such as the 'cloud'. What's more, the community has promptly readied the tools needed to meet this emerging trend. It is not surprising, then, that evangelists believe that open source has built the cloud...

We certainly do not need to tell you what free/open-source software is, but probably should spend a few minutes to clarify what cloud computing really is. Take any definition of cloud computing, and it sounds so similar to software-as-a-service, utility computing, and even grid computing, to boot. It takes quite a while to figure out the difference, which is why it is best explained to you right away, so you can appreciate the cloud computing concept and the open source advantage even better.

## Hey diddle diddle...

If the cow jumped over the moon today, what would it see on the clouds below? It would see software-as-a-service (SaaS), platform-as-a-service, utility computing, managed service providers, Web services, and cloud integrators. Well, that is what it is. Cloud computing is not a new technology; it is merely a new concept that integrates many virtualisation and pay-as-you-go models that already existed.

Software-as-a-service enables a user to rent and run software applications from service providers who maintain and manage the application on their servers. Remember Salesforce.com. The user simply has to rent an app, use it over the Web, and pay as per usage.

Utility computing, similarly, enables users to rent infrastructure, such as storage or servers, and use it over the Web. Companies use such services to cater to temporary surges in requirements. Remember IBM, Sun and Amazon.com.

Web services enable developers to connect or fit functional blocks or application programming interfaces (APIs) offered over the Web into their own applications, so as to not reinvent the wheel. Remember Google Maps and Xignite.

On-demand platforms enable users to string together whatever applications they need from a service provider into a seamless solution, and use it over the Internet. This is somewhat like an extension of SaaS, but here the user picks and uses a group of applications, rather than just one. The choice of applications is, however, limited by whatever the service provider has on offer. Remember Force.com, Coghead, and Google AppEngine.

Managed services is also a similar concept; it's just that the user is offered a service—such as network security or backup—over the Web, instead of just an application. Remember IBM, Symantec and Verizon.